

RMO(INDIA) – 2003

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be a triangle in which $AB = AC$ and $m\angle CAB = 90^\circ$. Suppose M and N are points on the hypotenuse BC such that $BM^2 + CN^2 = MN^2$. Prove that $m\angle MAN = 45^\circ$.
 2. If n is an integer greater than 7, prove that $\binom{n}{7} - \left\lfloor \frac{n}{7} \right\rfloor$ is divisible by 7. Here $\binom{n}{7} - \left\lfloor \frac{n}{7} \right\rfloor$ denotes the number of ways of choosing 7 objects from n objects and $[x]$ is greatest integer less than equal to x.
 3. Let a, b, c be three positive real numbers such that $a + b + c = 1$. Prove that among the three numbers $a - ab$, $b - bc$, $c - ca$ there is one which is at most $\frac{1}{4}$ and there is one which is at least $\frac{2}{9}$.
 4. Find the number of order triples (x, y, z) of non negative integers satisfying the conditions:
 - (a) $x \leq y \leq z$.
 - (b) $x + y + z \leq 100$.
 5. Suppose P is an interior point of a triangle ABC such that the ratios $\frac{d(A, BC)}{d(P, BC)} = \frac{d(B, CA)}{d(P, CA)} = \frac{d(C, AB)}{d(P, AB)}$. Find the common value of these ratios. Here d(X, YZ) denotes the perpendicular distance from X to line YZ.
 6. Find all real numbers 'a' for which the equation $x^2 + (a - 2)x + 1 = 3|x|$ has exactly three distinct real solutions in x.
 7. Consider the set $X = \{1, 2, 3, \dots, 10\}$. Find two disjoint nonempty subsets A and B of X such that:
 - (a) $A \cup B = X$.
 - (b) $\text{prod}(A)$ is divisible by $\text{prod}(B)$, where for any finite set of numbers C; $\text{prod}(C)$ denotes the product of all numbers in the set.
 - (c) The quotient $\text{prod}(A)/\text{prod}(B)$ is as small as possible.

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